

To keep this server running smoothly for months without Swap filling up or apps crashing again, apply these **four set-and-forget maintenance rules**:

1. Make swappiness=10 Permanent

Right now, your swap is sitting at 3.4 GiB because Linux's default behavior (swappiness=60) slowly pushes active memory into disk swap over weeks of uptime. Locking it to 10 forces Linux to use fast RAM first and only touch Swap during real emergencies:

```
Bash
# Check if it's already set permanently
cat /proc/sys/vm/swappiness

# If nothing prints, append it permanently:
echo "vm.swappiness=10" | sudo tee -a /etc/sysctl.conf
```

2. Add a Memory Safety Limit to PM2 (--max-memory-restart)

With 26 Node apps running 24/7, even a tiny 5 MB/day memory leak in one app will clog your RAM over 16+ days.

Tell PM2 to automatically and silently recycle any app that crosses a safe RAM threshold (e.g., **300 MB** for APIs/Frontends, **500 MB** for n8n):

```
Bash
# Example: restart an app only if it leaks beyond 300MB
pm2 start npm --name="consultbigdata-frontend" --max-memory-restart 300M -- start-p
Ctrl+C
pm2 save
```

(If you use an ecosystem.config.js file, add max_memory_restart: '300M' to each app's config block, then run pm2 reload all and pm2 save.)

3. Cap MariaDB's Memory Permanently

MariaDB will slowly grow its buffer cache over days until it takes up 30–40% of your RAM. Make sure its InnoDB buffer pool is permanently capped at **1 GB**:

1. Open the MySQL/MariaDB server config:

```
Bash
```

```
sudo nano /etc/mysql/mariadb.conf.d/50-server.cnf  
# (or /etc/my.cnf if that file doesn't exist)
```

2. Verify or add this under [mysqld]:

```
Ini, TOML
```

```
[mysqld]
```

```
innodb_buffer_pool_size = 1
```

3. Save and close (Ctrl + O, Enter, Ctrl + X).

4. Schedule a Weekly Nightly Maintenance Cron (Optional but Recommended)

To prevent stale pages from permanently getting trapped in Swap over months of uptime, add a lightweight cron job that clears filesystem cache once a week at 3:00 AM Sunday:

```
Bash
```

```
# Open crontab
```

```
sudo crontab -e
```

```
# Add this line at the bottom to flush cache weekly on Sunday at 3 AM:
```

```
3 * * 0 sync && echo |>/proc/sys/vm/drop_caches
```

Summary of Your Safe Operating Limits

- **Baseline Free RAM:** Should stay around **2.0 GB – 2.5 GB**.
- **Next.js Builds:** Always run with `NODE_OPTIONS="--max-old-space-size=1536"` npm run build so compilation never eats into the ~4.5 GB baseline used by your 26 apps.
- **Server Load Average:** Should hover below **2.0** under normal traffic on an 8 GB VPS.